

13.1 R Markdown: Chunks and Tables

Label chunks, set options, `kable()` a table, and include a child document.

1. Label every chunk

Note 1.4 covered Render, YAML, and `echo` / `eval` / `include`. This note is the next layer: names, defaults, tables, child files.

Give each chunk a unique **label**. Letters, digits, and hyphens are fine. The older fence `{r bill-summary}` is the same idea.

```
```${r}
#| label: bill-summary

penguins |> summarize(n = n())
```
```

2. Options that hide or skip work

| Option | What it hides or skips |
|----------------------|---|
| <code>eval</code> | If <code>false</code> , R does not run the chunk. The reader can still see the code. |
| <code>echo</code> | If <code>false</code> , the reader does not see the code. Results still appear. |
| <code>include</code> | If <code>false</code> , nothing from the chunk is printed. The code still runs if <code>eval</code> is true. |
| <code>warning</code> | If <code>false</code> , warnings are hidden. |
| <code>message</code> | If <code>false</code> , messages are hidden (<code>library()</code> chatter, <code>read_csv()</code> types). |
| <code>cache</code> | If <code>true</code> , knitr stores the result and skips a rerun when the chunk has not changed. |

`include: false` is the usual setup chunk: load packages, read data. Later chunks need those objects. The reader does not need that boilerplate.

`cache: true` is a speed trick. Turn it off for a final Render so you are not looking at a stale object.

3. Defaults in a setup chunk

Set the usual rules once with `knitr::opts_chunk$set()`. A later chunk can override one option.

```
```${r}
#| label: setup
#| include: false

knitr::opts_chunk$set(
 echo = TRUE,
 message = FALSE,
 warning = FALSE
)
```
```

Quarto's YAML `execute:` block (note 1.4) does the same job. Either is fine. This week we use `opts_chunk$set()` so the habit works in an old `.Rmd` as well as a `.qmd`.

4. Tables with `kable()`

`print()` of a tibble is readable in the console. In a report, pass the table to `knitr::kable()` and give it a caption.

```
penguins |>
  group_by(species) |>
  summarize(
    mean_bill = mean(bill_length_mm, na.rm = TRUE),
    n = n(),
    .groups = "drop"
  ) |>
  knitr::kable(caption = "Mean bill length (mm) by species.", digits = 1)
```

`digits` rounds numbers. `{kableExtra}` can style rows; we do not need it this week.

5. Folding code in the YAML

R Markdown:

```
output:
  html_document:
    code_folding: hide
```

Quarto:

```
format:
  html:
    code-fold: true
```

`hide` / `true` starts with the chunks collapsed. The reader can open one.

6. Child documents

A long report can pull in another file:

```
```{r, child = "helper.qmd"}```
```

The child's YAML is **ignored**. Title, author, and `format:` live only in the parent. The child is just Markdown plus chunks, run in the parent's session.

Labels must stay unique across parent and child. If a helper also has a chunk named `setup`, rename one of them.

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## 7. Load a package only if it is missing

When a child (or a chunk you rerun) would call `library(tidyverse)` a second time, skip it:

```
```{r, eval = !"tidyverse" %in% .packages()}```  
library(tidyverse)  
```
```

`.packages()` is the character vector of attached packages. The `!` flips the test: evaluate the chunk only when tidyverse is **not** already loaded.

The same idea for data: `eval = !exists("penguins")`.

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## 8. Practice

1. List four chunk options and say what each one hides (or skips).
2. Write a setup chunk that sets `echo = TRUE` and `message = FALSE` for the whole document.